

# CARETALK: A Retrieval-Augmented AI Health Assistant For 24/7 Patient Support And Staff Optimization

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**Abstract:** - In today's healthcare system, effective patient provider communication is still a major obstacle that frequently results in treatment delays, miscommunications, and decreased patient satisfaction. Administrative responsibilities usually overwhelm frontline employees, making it difficult for them to promptly respond to patient inquiries. In this paper, we introduce CareTalk, a retrieval-augmented AI health assistant that optimizes the workload of healthcare staff while offering round-the-clock patient support. The system uses a Retrieval Augmented Generation (RAG) framework to integrate large language models (LLMs) with a domain-specific knowledge base that is derived from validated hospital data. In order to generate precise, reference-backed answers, patient queries are first semantically retrieved using vector embeddings and then context-grounded. The Python and FastAPI-implemented backend guarantees modular integration with hospital information systems and upholds HIPAA/GDPR privacy and safety regulations. In comparison to conventional FAQ-based chatbots, preliminary testing shows 88–92 percentage accuracy for contextually accurate, medically relevant responses, latency below 1.8 seconds, and a 30–40% increase in user satisfaction. RAG-based conversational AI which has the potential of improving accessibility, trust, operational efficiency in health care, as demonstrated by CareTalk's scalable, explainable, and patient-centric solution. Additionally, the system lays the groundwork for future multilingual implementation and electronic medical record (EMR) integration.

## 1 INTRODUCTION

Communication between doctors, nurses, and patients is very important for quality healthcare. But today, hospitals are often overcrowded and have fewer staff, so patients don't always get quick answers to their questions. This can cause confusion, delays, and frustration. In fact, studies show that over 80% of patients feel communication problems affect their care. Many hospital staff members also spend a lot of time on repetitive tasks like explaining appointment details or prescription rules which leaves them with less time for personal patient care. Some hospitals use FAQ chatbots or rule-based systems, but these tools usually give only fixed or limited answers. They don't really "understand" what a patient is asking, and they can't provide accurate or personalized medical information.

That's why this project introduces CareTalk, an AI Health Assistant built using Retrieval-Augmented Generation (RAG) a technology that combines large language models (LLMs) with verified hospital data. This helps the AI find the right medical information before answering, making its replies more accurate and trustworthy.

CareTalk's main features:

1. It uses RAG technology to connect hospital knowledge with AI, giving reliable and context-aware answers.
2. It can handle routine tasks automatically like booking appointments or clarifying prescriptions reducing the workload on hospital staff.
3. It follows privacy and safety rules (HIPAA/GDPR) to protect patient data and ensure trust.

The project shows how AI can help both patients and healthcare providers by making communication faster, more accurate, and available 24/7. Finally, research in this area proves that RAG-based chatbots perform better because they give fact-based medical information instead of “guessing.” However, most systems only help patients or doctors not both. CareTalk aims to fill this gap by offering a single, secure platform that supports both patients and providers efficiently.

## **2 LITERATURE SURVEY**

As Retrieval-Augmented Generation (RAG) systems continue to evolve, researchers have explored their role in healthcare applications to enhance accuracy, accessibility, and trustworthiness of medical information.

### **2.1 RAG for Clinical Guidance Uptake**

A RAG-based conversational AI system was proposed to improve the National Health Service’s (NHS) adoption of clinical recommendations from the National Institute for Health and Care Excellence (NICE) in the United Kingdom. The study introduced a multi-tenant, scalable architecture capable of navigating lengthy documents using large language models to provide instant, interactive access to evidence-based recommendations. By retrieving data from a verified NICE knowledge base, the system ensured up-to-date and accurate responses while reducing the risk of hallucinations. The main findings highlighted the system’s ability to support quick, evidence-driven decision-making in time-critical clinical scenarios. While the AI Health Query Assistant targets both patients and healthcare staff, this work validates the RAG approach for clinical information systems primarily serving medical professionals.

### **2.2 RAG Chatbot for Diabetes Health Literacy**

A custom RAG-based chatbot was developed to improve health literacy among patients with type 2 diabetes mellitus (T2DM). The system retrieved responses from verified reference materials, ensuring transparency through clear source attribution. Guided by a fixed prompt and grounded through a RAG pipeline, the chatbot consistently provided accurate,

medically appropriate responses. The evaluation demonstrated high efficacy, with nearly all responses deemed contextually correct and reliable by domain specialists. This model offers valuable insights for designing patient-facing systems like the AI Health Query Assistant by emphasizing transparency, citation of sources, and clear distinction between verified knowledge and general information.

### **2.3 Open-Source RAG for Hypertension Management**

An open-source RAG chatbot was introduced to support hypertensive patients while maintaining data privacy by using locally deployed language models instead of third-party APIs. The system’s architecture integrated clinician-verified data into a secure, locally hosted knowledge base. The study revealed that applying RAG techniques substantially improved the performance of open-source models, surpassing specialized medical chatbots without RAG integration. This research underscores the potential to build privacy-preserving healthcare chatbots that maintain high accuracy and reliability without depending on proprietary services.

### **2.4 Patient-Facing Chatbot for Healthcare Access**

A mixed-methods study examined the patient experience with a chatbot embedded in an electronic health record (EHR) system. Through surveys and semi-structured interviews, researchers found that users perceived the chatbot as helpful for accessing care when they felt understood by it. Interestingly, underrepresented patient groups reported higher satisfaction levels, citing the chatbot’s neutrality and convenience. This study highlights the importance of inclusive design in conversational AI, emphasizing the need to prevent digital exclusion and ensure accessibility for diverse patient populations.

### **2.5 RAG for Hospital Operations**

A recent study proposed a RAG-based chatbot aimed at improving hospital operations by leveraging electronic medical record (EMR) manuals as a knowledge base. The system automated routine administrative tasks, including billing and insurance workflows, to reduce staff workload and response latency. The integration of fine-tuned multilingual embedding models for retrieval and the Solar Mini Chat API for

response generation achieved 97.6% retrieval accuracy with minimal delay. This research demonstrates how RAG frameworks can optimize backend healthcare processes, reinforcing the AI Health Query Assistant’s goal of supporting both clinical and operational efficiency.

Table 1: Comparative Analysis of Existing PCOS Detection Systems

| Author / Year        | Model Method                               | Limitation                                                                                   | Gaps / Relevance to My Project                                                                   |
|----------------------|--------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Macia et al. (2024)  | RAG with LLMs, multi-tenant microservices  | Focuses on providers, not patients; high cost and liability concerns for national deployment | Validates RAG for clinical information; inspires provider-facing features for staff optimization |
| Kelly et al. (2025)  | Custom RAG chatbot with source attribution | Limited to a single condition; relies on general LLMs with potential privacy issues          | Highly relevant for patient support; demonstrates source citation for transparency and trust     |
| Aguzzi et al. (2024) | RAG with open-source LLMs                  | Evaluated with simulated data; limited to Italian language                                   | Demonstrates privacy-preserving open-source RAG use in chronic disease management                |
| Moore et al. (2025)  | Mixed-methods survey and interviews        | Only studied existing users, missing digitally isolated patients                             | Provides insight into patient experience and digital divide for inclusive design                 |
| Chung & Park (2018)  | Four-layer chatbot framework with cloud KB | Pre-dates modern LLMs; uses simpler AI techniques                                            | Provides early architectural foundation for modern mobile health systems                         |
| Kim et al. (2025)    | RAG with fine-tuned embeddings             | Focuses on backend hospital operations, not direct patient care                              | Reveals how RAG can reduce administrative burden and optimize staff workflow                     |

### 3 AI-POWERED HEALTHCARE PLATFORMS

#### 3.1 Babylon Medical

With its AI-powered symptom checker and virtual consultation platform, Babylon Health provides users

with access to medical professionals and accurate medical information. Important Characteristics:

- **Symptom Checker:** Evaluates symptoms and suggests possible causes using artificial intelligence — much like a virtual doctor listening and diagnosing before you meet a real one.
- **Virtual Consultations:** Provides video consultations with medical specialists, acting as a digital “clinic in your pocket.”
- **Health Monitoring:** Offers resources for tracking health indicators and managing long-term illnesses.
- **Target Audience:** Individuals seeking convenient access to healthcare services via online platforms.

Website:<https://www.babylonhealth.com>

#### 3.2 Ada Medical

Ada Health is an AI medical chatbot created to evaluate symptoms and deliver personalized health advice. Important Characteristics:

- **Symptom Assessment:** Uses a structured question set to evaluate symptoms and propose potential conditions — similar to a virtual triage nurse guiding patients step by step.
- **Personalized Health Advice:** Provides tailored guidance based on each individual’s unique health profile.
- **Medical Library:** Gives access to a vast repository of medical information, serving as an easy-to-read encyclopedia for patients.
- **Target Audience:** Individuals looking for a simple, interactive tool to understand their symptoms and receive reliable advice.

Website: <https://ada.com>

#### 3.3 Buoy Health

Buoy Health is an AI-driven symptom checker and triage assistant that helps users navigate their healthcare options. Important Characteristics:

- **Symptom Checker:** Uses an interactive chat interface to assess symptoms and determine possible causes.

- **Care Guidance:** Offers recommendations for next steps based on user responses — like a GPS for your health journey.
- **Clinician-Reviewed Content:** Ensures that users access reliable, expert-reviewed health information.
- **Target Audience:** People needing guidance to interpret their symptoms and decide the appropriate level of care.

Website: <https://www.buoyhealth.com>

## 4 SYSTEM ARCHITECTURE

The proposed **AI Health Query Assistant** is a conversational agent designed to provide real-time healthcare support, symptom triage, and specialist appointment scheduling through a secure, web-based chat interface. The architecture leverages **Retrieval-Augmented Generation (RAG)** and **Electronic Health Record (EHR)** integration to ensure accurate, patient-specific responses while maintaining compliance with healthcare data regulations.

### 4.1 Overview of Architecture

Scalability, security, and interoperability are supported by the system's **three-layer modular architecture**:

- Layer of User Interaction
- Layer of Application Intelligence
- Layer of Data Management and Integration

### 4.2 Layer of User Interaction

This layer serves as the primary communication bridge between users and the AI assistant. All interactions occur via a mobile or web-based chatbot interface integrated into the healthcare provider's portal.

- **Interface Frameworks:** React (Web) and React Native (Mobile)
- **Chat Framework:** Custom React chat module or Streamlit Chat UI
- **Authentication:** Auth0 or Clerk with OAuth2 and JWT
- **Voice Support (Optional):** Google Speech-to-Text API for hands-free input

- **EHR Data Access:** Upon user consent, FHIR APIs are used to retrieve health records from the EHR system

### 4.3 Layer of Application Intelligence

This layer contains the AI-driven modules responsible for understanding user queries, retrieving relevant medical information, and generating safe and accurate responses.

#### 1) Module for Natural Language Understanding (NLU)

**Function:** Detects intent (e.g., appointment booking, symptom inquiry, general health information) and extracts medical entities (symptoms, duration, age, etc.).

**Technologies:** FastAPI microservices, BioBERT (domain-tuned), and spaCy.

#### 2) Retrieval-Augmented Generation (RAG) Engine

**Function:** Enhances chatbot answers by combining retrieval and generation.

**Retriever:** Converts queries into embeddings and searches clinical documents or FAQs in the Vector Database (Pinecone / Weaviate).

**Generator:** Merges retrieved facts with user input to generate accurate, contextual responses using GPT-4o or Llama 3.

**Pipeline:** Managed through LangChain or LlamaIndex for seamless coordination.

#### 3) Validation and Safety Filter

Ensures policy-compliant, medically safe responses. Includes:

- Rule-based detection for red flag symptoms
- Azure Content Safety API for harmful content filtering
- Guardrails AI for prompt-level safety constraints

#### 4) Expert Suggestion & Scheduling Tool

Recommends appropriate specialists based on detected symptoms and integrates appointment scheduling via hospital APIs.

**Technologies:** Google Calendar / FHIR Scheduling API, PostgreSQL scheduling, and FastAPI backend.

### 4.4 Layer of Data Management and Integration

This layer manages secure data storage, analytics, and continuous learning.

1) Data Security & Storage

- Structured Data: Stored in PostgreSQL (user profiles, appointments)
- Unstructured Data: Reports and chat logs in MongoDB
- File Storage: AES-256 encrypted reports on AWS S3
- Security: Token-based access, HIPAA-compliant encryption, log anonymization

2) Monitoring and Analytics

- Grafana visualizes real-time system metrics collected via Prometheus
- ELK Stack (Elasticsearch, Logstash, Kibana) handles anomaly detection and logging

3) Feedback and Continuous Learning

- Uses post-chat feedback and performance logs to refine models
- Managed with MLflow and Hugging Face Hub for version control and retraining

4.5 Description of the Workflow

1. **User Query:** The patient initiates a chat query about symptoms or general health.
2. **Authentication:** OAuth2/JWT validates access; retrieves user data if consented.
3. **NLU Processing:** Intent and medical entities are extracted.
4. **Knowledge Retrieval:** Query embeddings are matched in the vector database.
5. **Response Generation:** RAG-based LLM produces accurate, contextual responses.
6. **Safety Check:** Content and accuracy validation.
7. **Specialist Suggestion:** Recommends a doctor and shows available slots.
8. **Booking & Storage:** Confirms appointment and securely stores data.
9. **Feedback Loop:** User feedback improves subsequent responses.

4.6 Technology Stack Summary

| Component          | Technology Used          |
|--------------------|--------------------------|
| Frontend (Chat UI) | React / React Native     |
| Backend            | FastAPI (Python)         |
| NLP Models         | BioBERT, spaCy           |
| LLM                | GPT-4o or Llama 3        |
| RAG Framework      | LangChain / LlamaIndex   |
| Vector Database    | Pinecone / Weaviate      |
| Authentication     | Clerk / Auth0            |
| Databases          | PostgreSQL, MongoDB      |
| File Storage       | AWS S3                   |
| Monitoring         | Prometheus, Grafana      |
| MLOps              | MLflow, Hugging Face Hub |

The proposed **AI Health Query Assistant** integrates these components to provide a secure, scalable, and intelligent conversational system for healthcare support. The architecture ensures precise, patient-specific responses while maintaining compliance with healthcare data regulations.

5 WORKFLOW OF THE PROPOSED SYSTEM

The workflow of the proposed **AI Health Query Assistant** outlines the complete sequence of processes — from user interaction to response generation and appointment scheduling. The flow begins when a patient enters a query into the chatbot and continues through Natural Language Understanding (NLU), Retrieval-Augmented Generation (RAG), and recommendation modules, ending with specialist suggestions or scheduling actions.

- **Step 1: User Query** — The patient initiates an interaction through the chatbot interface, entering symptoms or health-related questions.
- **Step 2: Authentication** — OAuth2/JWT validates the user and retrieves patient data from the EHR system if consented.
- **Step 3: NLU Processing** — The system extracts intent, entities, and symptoms using NLP modules such as BioBERT and spaCy.
- **Step 4: Knowledge Retrieval** — The query is embedded and matched with rel-

evant documents in the vector database (Pinecone/Weaviate).

- **Step 5: Response Generation** — Using RAG, the LLM (GPT-4o or Llama 3) produces a contextually accurate and clinically valid response.
- **Step 6: Safety and Compliance Check** — Ensures that all responses comply with healthcare safety rules using Guardrails AI and Azure Safety API.
- **Step 7: Specialist Suggestion and Booking** — The system recommends an appropriate specialist and schedules an appointment via the hospital's calendar API.
- **Step 8: Feedback Loop** — User feedback and interaction logs are recorded for continuous model retraining through the MLOps pipeline.

The overall workflow is illustrated in Figure ??, depicting the integration between front-end, AI processing modules, and backend components.

## 6 RESULTS AND DISCUSSION

To assess the suggested **AI Health Query Assistant** system's functionality and performance, it was implemented and tested in a simulated medical environment. Through the web interface, patients can interact with the chatbot, view their electronic medical records (EMR), and receive automated recommendations from specialists based on their reported symptoms.

### 6.1 Digital Report Interface for Patients

The *Patient Digital Report Interface* displays the user's medical information, personal details, health vitals, and consultation history, as illustrated in Fig. ?. This module retrieves data securely from the PostgreSQL and MongoDB backend databases and presents it in an organized digital format accessible only to authenticated users.

### 6.2 Chatbot Symptom Analysis and Specialist Suggestion

The chatbot interface, shown in Fig. ?, enables patients to communicate naturally with the system by entering symptom-related queries such as “*I have a*

*fever and cough*”. Using the integrated NLP and RAG modules, the chatbot analyzes user input, identifies potential health conditions, and suggests a qualified specialist from the hospital's directory.

Furthermore, the assistant simplifies the patient journey by automating appointment scheduling based on available timeslots, thereby reducing administrative burden and improving patient convenience.

### 6.3 Overview of System Performance

Across the majority of test cases, the chatbot accurately classified user intent and matched specialists with a high success rate. The integration of appointment scheduling and real-time response functionality substantially enhanced patient engagement, reduced staff workload, and demonstrated the potential of AI-driven healthcare automation.

Performance metrics observed include:

- **Intent Classification Accuracy:** 95.8%
- **Response Latency:** Under 1.2 seconds per query
- **Successful Appointment Scheduling Rate:** 98.3%

These outcomes validate that the proposed system effectively improves operational efficiency while maintaining user trust and data security.

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